Fiscal Year:	FY 2010	**Task Last Updated:**	FY 09/14/2010
PI Name:	Brainard, George C. Ph.D.		
Project Title:	Optimizing Light for Long Duration Space Exploration		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Human Factors and Performance Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Philadelphia	**State:**	PA
Zip Code:	19107-5083	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Crew Health NNJ07ZSA002N
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lockley, Steven (Brigham and Women's Hospital) Hanifin, John (Thomas Jefferson University) Pineda, Maria (Thomas Jefferson University)		
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Task Description:

The goal of this project is to determine the optimal lighting for use as a countermeasure to sleep and circadian disruption for astronauts. Sleep deficits that diminish alertness, cognitive ability, and psychomotor performance are a serious risk factor during space missions. Currently, over 45% of all medications taken in space are sleep aids (Putcha et al., 1999). Despite their use, more than half of the astronauts still get only between 3 and 6 hours of sleep per night during spaceflight (Barger et al., 2009).
Light is the primary stimulus for human circadian regulation. On Earth, humans have a 24 h day/night cycle to maintain healthy circadian entrainment. In space, astronauts must contend with either rapidly changing or severely disrupted day/night cycles and must work in a dimly lit spacecraft interior with few windows. Bright white light has been implemented as a pre-launch countermeasure but has yet to be used during spaceflight. Providing sufficient light intensities to work areas in the ISS as well as future vehicles and habitats raises several concerns, among which are heat production, energy consumption, and up-mass. Improving these factors requires a better understanding of how different light sources regulate the human circadian system. From these data it may be possible to optimize astronaut and ground crew light exposure both as a stimulus for vision as well as a countermeasure for sleep and circadian disruption during space missions.

Currently, NASA uses white fluorescent light for interior illumination of both the STS and the ISS at relatively low intensities, and for a pre-launch lighting countermeasure for circadian disruption at much higher intensities. In the previous funding period, our NSBRI research was supported to determine if those fluorescent lamps have increased efficacy for melatonin suppression and circadian phase-shifting when they are enriched in the blue portion of the spectrum. Philips Lighting, an NSBRI industry partner, provided prototype blue-enriched lamps and exposure systems for that study. In addition, we began characterizing the circadian potency of ambient extraterrestrial light that astronauts will encounter on the surfaces of the Moon and Mars.

To develop a lighting countermeasure for space exploration, it is vital to determine the sensitivity of astronauts' circadian systems to both ambient and artificial lighting stimuli. In the current period of research, we are shifting our efforts from testing fluorescent light sources to testing solid-state lights. Artificial illumination for future space vehicles and habitats will be provided predominantly by light emitting diodes (LEDs). The following list identifies each specific aim and this past year's progress.

1. a. Determine the neuroendocrine, alerting, and circadian potency of light emitted by the white LEDs being considered for re-lamping the ISS as well as lighting future vehicles and habitats. Test the hypothesis that white LED light with a correlated color temperature of 6,500 K has increased efficacy for suppressing melatonin, phase-shifting, and enhancing alertness compared to NASA's current fluorescent lighting. For this aim, we worked with Apollo Health, to fabricate and install two 4' by 4' white LED lighting systems in our laboratory. To date, 8 healthy subjects have entered into the study and completed over 80 nighttime studies. Recently, The Lighting Sciences Group provided two different 4' by 4' white LED lighting systems to our lab. These systems emit white light a CCT of 4,000 K. Preliminary physical assessment of these new lighting systems is underway.

1. b. Assess the ocular safety of experimental subjects and astronauts exposed to light from LEDs at higher intensity levels than those currently specified for visual stimulation in space habitats. Working from national and international safety guidelines, a hazard analysis on the light emitted by our 6,500 K LED exposure systems has been completed confirming that these panels emit safe light exposures.

2. Determine the acute neuroendocrine, alerting, and performance enhancing potency of the ambient extraterrestrial light as filtered through spacesuit visors or glazed windows for Moon and Mars missions. Test the hypothesis that ambient lunar and Martian light, as filtered through spacesuit visors, will suppress melatonin less and will be weaker for enhancing alertness and performance compared to ambient daylight on Earth. For this aim, additional optical elements have been developed and installed for simulating Earth daylight in a Ganzfeld dome. To date, 8 healthy subjects have entered into the Ganzfeld Earth daylight study and completed over 60 nighttime studies.

Continuing prior work involves analysis of data from our 7-day phase shift study with fluorescent light at 17,000 K and 4,000 K. We have now calculated the acute melatonin suppression, circadian phase shifting, and subjective alerting capacity of these lamps in this study. Further analysis of neurobehavioral effects of these light sources is ongoing.

Together, the research from the two prior funding cycles along with the studies performed in the past year provide significant progress towards the development of lighting countermeasures for sleep and circadian disruption in ground crew members and astronauts. The results will also impact the NASA Human Integration Design Handbook and the Space Flight Human Systems Standard, NASA-STD-3001 that provide guidance for crew health, habitability, environment, and human factors in human spaceflight. Our progress addresses Critical Risk areas 13, 33, and 76 in the NASA Human Research Program Integrated Risk Plan (2009), and areas 26, 27, and 44 in the Bioastronautics Roadmap (2005). These areas concern countermeasures that will optimally mitigate performance problems associated with sleep loss and circadian disturbances and the "mismatch between crew physical capabilities and task demands."

Rationale for HRP Directed Research:

The knowledge we hope to gain from this research, though focused on spaceflight, will also benefit people on Earth. The sleep deficits experienced by astronauts during spaceflight can be considered a threat to the success of space missions (Bioastronautics Roadmap, Longnecker and Molins, 2005; NASA Human Research Program Integrated Risk Plan, 2009). The resulting physiological and behavioral changes caused by sleep and circadian disruption can lead to diminished alertness, cognitive ability and psychomotor performance (Dijk et al., 2001). As a measure to counteract sleep disruptions, over 45% of all medications taken in space are sleep aids (Putcha et al., 1999). A significant portion of the global population suffers from chronic sleep loss and/or circadian-related disorders. Evidence for disease or illness occurring due to a disruption of circadian homeostasis has mounted significantly in the past several years. In the United States, nearly 22 million Americans do shift work that interferes with a biologically healthy nocturnal sleep cycle (US Bureau of Labor Statistics, 2007). Shift workers have been shown to be more likely to suffer from a wide variety of ailments, including cardiovascular disease, gastrointestinal distress, and cognitive and emotional problems. Furthermore, epidemiological studies of female shift workers have shown that they are statistically more likely to suffer from breast and colon cancer as compared to their day shift counterparts. The World Health Organization has identified shift work as a probable risk for cancer (The International Agency for Research on Cancer, 2007). Our laboratory is involved in testing the hypothesis that nighttime exposure to light suppresses melatonin and contributes to cancer risk (Blask et al., 2005; Stevens et al., 2007). Aside from evidence of a breakdown in physical health, the effects of circadian disruption

Research Impact/Earth Benefits:	and sleep loss are known to have potentially dangerous behavioral effects. Mental fatigue, diminished alertness, loss of psychomotor coordination and decreased physical performance are all commonly found in individuals with sleep loss, sleep debt, or circadian misalignment. The impact of these dangers affects many industries, including transportation, manufacturing, communications, and medicine. It has long been a source of concern for the military and, more recently, homeland security. Many people also experience the same effects after air travel across several time zones. In the past, the U.S. Air Force has supported our laboratory to study the acute alerting effects of light (French et al., 1990; Brainard et al., 1996). Our current work for NIH has continued this effort (Lockley et al., 2006). Existing light therapy interventions stand to benefit from enhancing our understanding of how different wavelengths of the spectrum affect human circadian and neurobehavioral regulation. The result may be a more efficient intervention with increased potency and/or fewer side effects. One such disorder currently being treated with bright white light is Seasonal Affective Disorder (SAD), also known as winter depression. It is estimated that as many as 1 in 5 Americans suffer from SAD or its milder version, subsyndromal Seasonal Affective Disorder (sSAD) (Lam and Levitt, 1999). Similar bright white light interventions are also used to treat jetlag. Side effects from exposure to bright white light for these and other therapies include: hypomania, headache, vision problems, nausea, dizziness, and anxiety. Optimizing the light spectrum for specific affective and/or circadian-related disorders may deliver the same medical benefit with lower levels of light intensity, and potentially fewer side effects. Our group has completed Phase I testing of a lower intensity of light therapy with blue solid-state lighting for SAD patients (Glickman et al., 2006).
Task Progress:	A major goal of our work is to optimize lighting that supports vision and serves as an effective circadian countermeasure for astronauts and ground crew during space missions. In this past year, progress has been made on several of the new specific aims of our project. Two of our aims are to determine the ocular safety and neuroendocrine potency of light emitted by the white LEDs being considered by NASA for in-flight and habitat lighting systems. Although illumination from the LEDs appears to be white to the visual system, it is significantly richer in blue wavelengths compared to the white fluorescent light currently used by NASA. Working with Apollo Health, we designed, fabricated, and installed two 4' by 4' white LED lighting systems (6,500 K) in our laboratory. To assess the ocular safety of experimental subjects and astronauts exposed to light from the LEDs at intensity levels higher than those currently specified for visual stimulation in space vehicles and habitats, we worked with a medical physicist who has completed a hazard analysis, based on national and international safety guidelines, of the light emitted by our white LED exposure systems. The 6,500 K panels were confirmed as providing safe light exposures. To test the hypothesis that white LED light has increased efficacy for melatonin suppression compared to white fluorescent light, 8 healthy subjects were entered into a melatonin suppression study and have completed over 80 nighttime studies. Recently, two different 4' by 4' white LED lighting systems have been designed, fabricated, and sent to our lab by The Lighting Sciences Group. These systems emit white light with a correlated color temperature (CCT) of 4,000 K. Preliminary physical assessment of these new lighting systems is underway. A third specific aim is to compare the potency of the ambient light on the surface of Earth, Moon, and Mars for acute melatonin suppression, since circadian and sleep regulation depends on an astronaut's total exposure to both artificial light and passive ambient illumination, In the last funding period, we worked with optical engineers to create a Ganzfeld lighting system to simulate extraterrestrial light that astronauts would see on the surface of Moon and Mars as well as simulated Earth skylight. This equipment then was used for testing the potency of these light stimuli for their capacity to suppress melatonin in healthy men and women. In this past year, more than 60 new nighttime studies have been completed with simulated Moon, Mars, and Earth light. Continuing work from the previous funding cycle involves analysis of circadian and neurobehavioral data from the 7-day phase shift study with polychromatic fluorescent light at 17,000 K and 4,000 K. We have now calculated the acute melatonin suppression, circadian phase shifting, and subjective alerting capacity of these lamps in this study. Further analysis of neurobehavioral effects of these light sources is ongoing.
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